[Aim : 100/100 in maths]

आरंभ

ARITHMETIC PROGRESSION

Lecture #3

ATK[2] :-

General form of AP :- $a, a+d, a+2d, a+3d - - - - \boxed{a+(n-1)d}$

$a_1 \quad a_2 \quad a_3 \quad a_4 \; . - - - - a_n$

$$\boxed{a_n = a+(n-1)d}$$

? ?

#LP: If the 10th term of an A.P. is 52 and 17th term is 20 more than the 13th term, find the A.P. [CBSE 2006]

given $\boxed{a_{10} = 52}$

$\boxed{a + 9d = 52}$ — (I)

given $\boxed{a_{17} = 20 + a_{13}}$

$\cancel{a} + 16d = 20 + \cancel{a} + 12d$

$16d - 12d = 20$

$\cancel{4}d = \cancel{20}\,5$

$\boxed{d = 5}$

put

$a + 9(5) = 52$

$a + 45 = 52$

$a = 52 - 45$

$\boxed{a = 7}$

A.P. ?

↓

$a, a+d, a+2d,$

$a+3d \ldots$

$a = 7 \,,\; d = 5$

$AP \longrightarrow \quad a \,,\; a+d \,,\; a+2d \,,\; a+3d \; - \; . \; -$

$\quad 7 \,,\; 7+5 \,,\; 7+2(5) \,,\; 7+3(5) \; - \; -$

$$\boxed{AP: \quad 7 \,,\; 12 \,,\; 17 \,,\; 22 \; - \; - \; -}$$

#LP: The first term of an A.P. is 5, the common difference is 3 and the last term is 80 ; find the number of terms.

$a = 5$

$d = 3$

last term = 80

i.e. $a_n = 80$

$a + (n-1)d = 80$

$5 + (n-1)(3) = 80$

$(n-1)(3) = 80 - 5$

$(n-1)(\cancel{3}) = \cancel{75}\,25$

$n - 1 = 25$

$\boxed{n = 26}$

∴ 26 terms are there

#LP: Which term of the sequence −1, 3, 7, 11, ... is 95?

Sequence: $-1, 3, 7, 11 \ldots 95$

$a_1 \quad a_2 \quad a_3 \quad a_4$

Checking for AP

$d_1 = a_2 - a_1 = 3-(-1) \Rightarrow 4$

$d_2 = 7 - 3 \Rightarrow 4$

$d_3 = 4$

all c.d are same

Yes, it's AP

Now, AP: $-1, 3, 7, 11 \ldots 95$

Let n^{th} term is 95

$\therefore a_n = 95$

$a + (n-1)d = 95$

$-1 + (n-1)(4) = 95$

$(n-1)(4) = 95+1$

$(n-1)(4) = 96$

$n-1 = \frac{\cancel{96}^{24}}{\cancel{4}}$

$n-1 = 24$

$\boxed{n = 25}$ Ans

$\therefore 25^{th}$ term is 95.

#LP: Which term of the sequence 4 , 9 , 14 , 19 , ... is 124 ?

AP or not? Yes

AP : 4, 9, 14, 19 - - - - 124 ?

Let n^{th} term is 124

$\therefore a_n = 124$

$a + (n-1)d = 124$

$4 + (n-1)(5) = 124$

$(n-1)(\cancel{5}) = \cancel{120}\ 24$

$\boxed{n = 25}$

$\therefore$ $\boxed{25^{th} \text{ term is } 124.}$

कितनी terms

#LP: How many terms are there in the sequence 3 , 6 , 9 , 12 , ... 111 ?

n terms

AP : 3, 6, 9, 12 - - - - 111

a_n

AP check ?

Yes

Let n terms are there

$\therefore a_n = 111$

$a + (n-1)d = 111$

✓ ? ✓

$3 + (n-1)(3) = 111$

$(n-1)\cancel{(3)} = \cancel{108}\ 36$

$n-1 = 36$

$\boxed{n = 37}$

$\therefore$ 37 term are there in this sequence

#LP: Which term of the A.P. 121, 117, 113, ... is its first negative term?

HW

#LP: Which term of the arithmetic progression 5, 15, 25, ... will be 130 more than its 31^{st} term ? [CBSE 2006, 2017]

HW

#LP: Is -150 a term of the A.P. 11 , 8 , 5 , 2 , ... ?

HW

#LP: Is 184 a term of the sequence 3 , 7 , 11 , ... ?

HW

#LP: Determine the 10^{th} term from the end of the A.P. 4 , 9 , 14 , ….. , 254.

HW

#LP: Find the 8^{th} term from the end of the A.P. 7 , 10 , 13 , ... , 184

[CBSE 2005]

HW question

THANK YOU COODIESS !!